

End-to-End Supervised Learning Pipeline

Course : Machine Learning

Dataset : UCI Heart Disease Dataset

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Assignment Objective :

The objective of the assignment is to build a complete supervised learning workflow starting from raw data preprocessing to training and comparing multiple classifiers including Decision Trees, Rule based models, Knn and Ensemble Learning methods.

1. Dataset Description

The UCI Heart Disease Dataset is chosen for the assignment. The dataset contains medical records of 920 patients collected from four hospitals: Cleveland, Hungary, Switzerland, and VA Long Beach. The goal of the assignment is to predict whether a patient has heart disease or not. This is a binary classification problem where 0 means No Disease and 1 means Disease.

The dataset contains 13 input features.

- i. **Continuous numeric features :** Age, resting blood pressure (trestbps), serum cholesterol (chol), maximum heart rate achieved (thalch), and ST depression induced by exercise (oldpeak).
- ii. **Discrete numeric feature :** The number of major vessels colored by fluoroscopy (ca).
- iii. **Nominal categorical features :** Chest pain type (cp), sex, fasting blood sugar (fbs), resting ECG results (restecg), exercise induced angina (exang), and thalassemia type (thal).
- iv. **Ordinal feature :** The slope of the peak exercise ST segment is an ordinal feature.

The original target column had severity values from 0 to 4, but it was later binarized i.e., 0 for no disease and 1 for any level of disease. The class distribution is balanced with approximately 46 percent of patients having no disease and 54 percent having disease, which reduces the risk of class imbalance affecting model performance.

The correlation heatmap shows that thalch (maximum heart rate) has a notable negative correlation with the target, meaning patients with higher heart rate tend to be healthier. The ca and oldpeak features show positive correlation with disease.



Figure 1 - Target class distribution

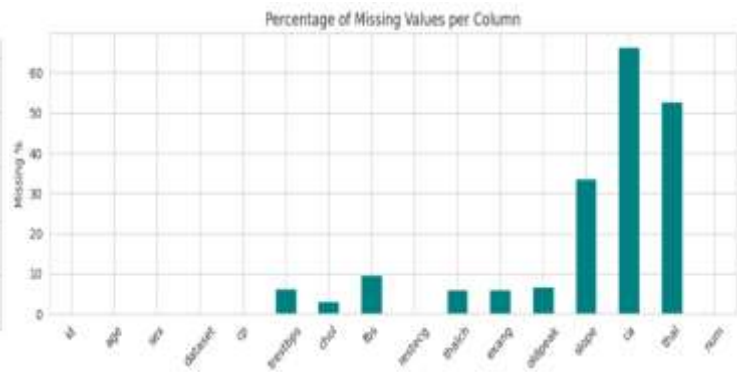


Figure 2 - % of Missing values per column

2. Data Preprocessing

A structured preprocessing pipeline was applied to the raw data before any model training. Each step was carefully chosen to ensure data quality and prevent information leakage from the test set into training.

- i. **Handling Missing Values:** Several columns including ca and thal had a significant percentage of missing entries.
 - Numeric columns were filled using the median value because it is more robust to outliers compared to the mean.
 - Categorical columns were filled using the most frequent value (mode). This preserves the natural distribution of the data without introducing extreme values that could distort model learning.
- ii. **Removing Irrelevant Columns:** The "id" and "dataset" columns were dropped as they carry no predictive value.
- iii. **Duplicates:** Checked and removed if any duplicate rows were found.
- iv. **Target Binarization:** The original target "num" has values 0-4. Converted to binary: 0 = No Disease, 1 = Disease.
- v. **Encoding Categorical Variables:** All nominal columns were converted using One-Hot Encoding. This creates a separate binary column for each category. The drop_first=True parameter was used to remove one column per group, avoiding the dummy variable trap which causes multicollinearity.

Label Encoding was deliberately avoided for nominal features because it would incorrectly imply a numeric ordering between categories that does not exist. For example, treating asymptomatic chest pain as mathematically greater than typical angina.
- vi. **Feature Scaling:** StandardScaler was applied after splitting to bring all features to mean zero and standard deviation one. This step is critical for distance-based models like kNN and gradient-based models like Logistic Regression. Without scaling, a feature like cholesterol with a range of 100 to 400 would unfairly dominate over age with a range of 0 to 80.

The scaler was fitted only on the training set and then applied to the test set, strictly preventing any information from the test data leaking into the training process.

- vii. **Train-Test Split:** The dataset was divided into 80 percent for training (736 samples) and 20 percent for testing (184 samples) using stratified sampling, which ensures the same class proportion is maintained in both splits.

3. Model Training and Evaluation

Task B - Baseline Models: Two baseline classifiers were trained first to set a performance benchmark.

- **Logistic Regression :** It finds a linear decision boundary by estimating the probability of heart disease using the sigmoid function. It is interpretable through its feature coefficients and achieved an accuracy of 0.8424 with a ROC-AUC of 0.9032, showing good class separation ability.
- **Naive Bayes :** It applies Bayes theorem under the assumption that all features are statistically independent of each other. Despite this simplifying assumption which rarely holds in real medical data, it matched Logistic Regression at 0.8424 accuracy with a slightly lower ROC-AUC of 0.8869, demonstrating its effectiveness as a fast baseline

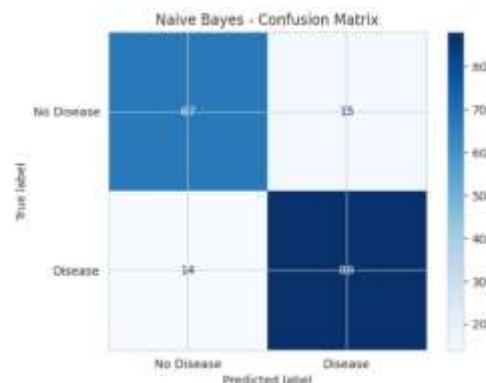
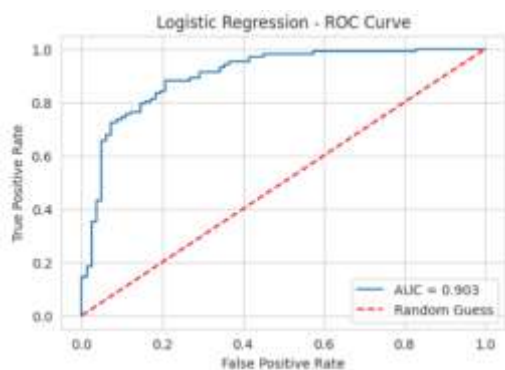


Figure 3-Logistic Regression ROC Curve **Figure 4-**Naïve Bayes Confusion matrix

Task C - Decision Tree: A Decision Tree Classifier was trained and tuned across three hyperparameters.

- The `max_depth` was tested from 1 to 14. At shallow depths, both train and test accuracy were low due to underfitting. At deeper levels, training accuracy climbed to near 100 percent while test accuracy dropped, showing classic overfitting behaviour. A depth of 5 was selected as the best balance.
- The `min_samples_split` was tested at values 2, 5, 10, and 20, controlling the minimum samples required before a node can split.

- The `min_samples_leaf` was tested at 1, 5, 10, and 15, controlling the minimum samples required in each leaf node.

These two parameters prevent the tree from forming tiny, noisy branches that memorise specific training samples. The final model achieved accuracy 0.7717.

Feature importance analysis showed that asymptomatic chest pain (`cp_asymptomatic`) was the single most important predictor, followed by maximum heart rate (`thalch`) and ST depression (`oldpeak`), all of which are clinically recognised heart disease indicators.

Task D - Rule-Based Classification: A shallow Decision Tree (`max_depth=4`) was trained to extract human-readable IF-THEN rules. This approach (Option 1) extracts rules directly from the tree structure. The model produced 8 rules with accuracy of 0.7935.

One example rule predicts No Disease with 80 percent confidence when chest pain is not asymptomatic and max heart rate is above a threshold. Another predicts Disease with 88 percent confidence when asymptomatic chest pain is present and `oldpeak` is high.

Interpretability vs Performance: The shallow rule-based model (accuracy 0.7935) is easier to understand than the deeper Decision Tree (accuracy 0.7717 at depth 5). In medical settings, doctors prefer interpretable rules even at the cost of a small accuracy reduction, because they need to understand and trust predictions before acting.

Task E - kNN Lazy Learning: The k-Nearest Neighbors classifier was tested with k values of 1, 3, 5, 7, and 9 using Euclidean, Manhattan, and Chebyshev distance metrics. kNN is called a lazy learner because it stores all training data and performs the entire computation at prediction time by finding the k most similar patients by distance and taking a majority vote.

Feature scaling is essential for kNN. Without it, cholesterol (100-400) would dominate distance calculations over age (0-80), making similarity unfair. After `StandardScaler` all features contribute equally to the distance.

- $k=1$ showed classic overfitting i.e., perfect training accuracy but test accuracy of only 0.8261.
- $k=5$ gave the best test accuracy of 0.8533 with a healthy bias-variance balance.
- Beyond $k=5$ accuracy declined due to underfitting as the model averaged too many neighbours.

Euclidean distance performed marginally best among the three metrics tested.

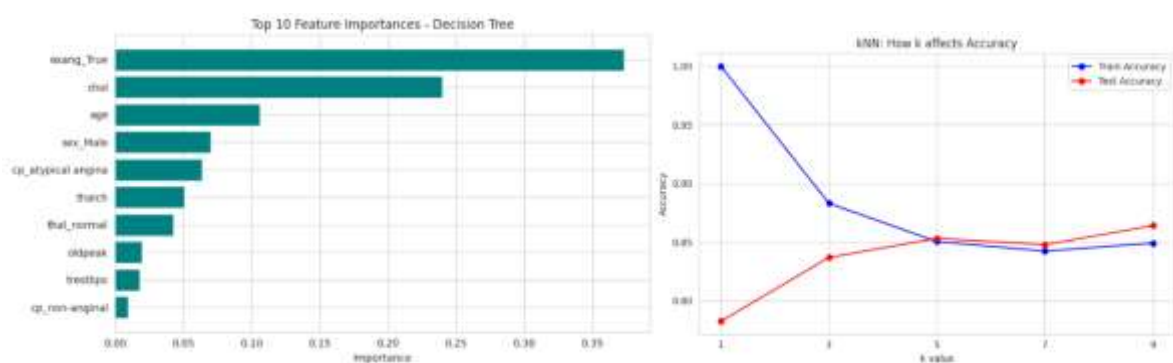


Figure 5-Decision Tree Top Feature Importance

Figure 6- Knn (K value vs Accuracy)

Task F - Ensemble Learning: Three ensemble models were trained.

- **Random Forest** : It builds 100 independent trees on random data subsets and combines predictions by majority vote (bagging), reducing variance.
- **Gradient Boosting** : It trains trees sequentially where each corrects the previous ones errors (boosting), reducing bias.
- **AdaBoost** : This model also boosts but increases weight on misclassified samples each round.

Random Forest achieved the highest accuracy of 0.8641 and ROC-AUC of 0.9207. AdaBoost tied kNN at 0.8533. All ensemble models outperformed the single Decision Tree at 0.7717, confirming that combining learners reduces overfitting.

The tradeoff is that ensemble models act as black boxes and cannot easily explain individual predictions, reducing clinical trust.

Task G - Final Comparison and Recommendation : All eight classifiers were evaluated on the same held-out test set of 184 samples. The bar chart below compares accuracy, F1-score, and ROC-AUC across all models, while the ROC curves show how well each model separates disease from no-disease patients across all decision thresholds.

Random Forest leads on all three metrics with the highest accuracy and AUC. Logistic Regression shows a strong AUC of 0.9032 despite lower accuracy, meaning it ranks disease patients well. Decision Tree has the lowest AUC of 0.8575

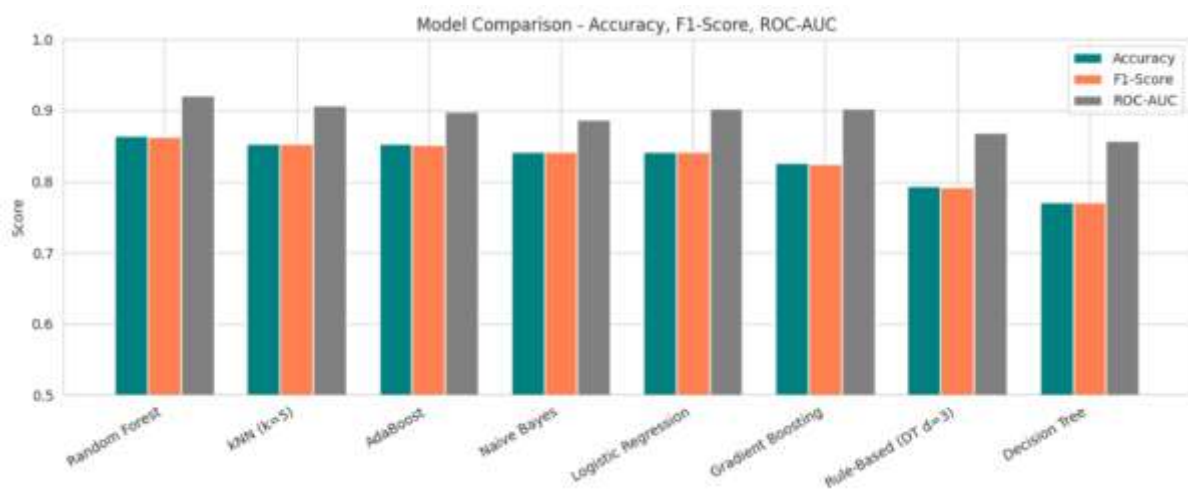


Figure 7 – Model comparison

4. Analysis and Recommendations

Q1. Which model performed best and why?

Random Forest performed best with Accuracy = 0.8641, F1-Score = 0.8631, and ROC-AUC = 0.9207. It outperformed all other models because it combines 100 independent Decision Trees and takes a majority vote. This approach reduces the overfitting problem

that a single Decision Tree suffers from. Random Forest also handles noisy medical data well because it trains each tree on a random subset of features and samples, making it robust to outliers and irrelevant features. The high ROC-AUC of 0.9207 confirms it separates disease and no-disease patients very effectively.

Q2. Which model is most interpretable and why?

The Rule-Based model (shallow Decision Tree with `max_depth=4`) is the most interpretable. It produces simple IF-THEN rules that any doctor or non-technical person can read, understand, and verify manually. For example: IF chest pain is asymptomatic AND max heart rate is low THEN → Disease. Logistic Regression is also interpretable through its coefficients. In contrast, ensemble models like Random Forest use 100+ trees and act as a black box i.e., they perform well but cannot easily explain individual predictions.

Q3. If this model was deployed in real life, what risks exist?

- **False Negatives:** If the model predicts No Disease but the patient actually has it, this could lead to a missed diagnosis i.e., which is life-threatening in heart disease cases.
- **Data Bias:** The dataset was collected from US and European hospitals. Predictions may be less accurate for patients from other regions like India due to differences in lifestyle, diet, and genetics.
- **Class Imbalance:** If the training data had significantly more healthy patients, the model would learn to predict No Disease more often, reducing sensitivity.
- **Black Box Problem:** Ensemble models like Random Forest are difficult to explain to doctors and patients. In medical AI, lack of explainability reduces trust and adoption.
- **Model Staleness:** Patient patterns and medical standards change over time. A model trained today may become less accurate over years if not regularly retrained.

5. Conclusion

This assignment built a complete end-to-end supervised learning pipeline on the UCI Heart Disease Dataset.

Random Forest achieved the best predictive performance with accuracy 0.8641 and ROC-AUC 0.9207. The Rule-Based Decision Tree offered the most interpretable output through clear IF-THEN rules at accuracy 0.7935. Asymptomatic chest pain type, maximum heart rate, and ST depression were consistently the strongest predictors of heart disease across all models.

The most important takeaway is that there is no universally best model. The right choice always depends on the context. Ensemble methods deliver the highest accuracy but sacrifice interpretability. Rule-based and logistic models are more transparent and trustworthy in clinical settings even at a small accuracy cost. In a real hospital deployment, the ideal solution would pair a high-performing ensemble model with an explainability framework, producing predictions that are both accurate and understandable to the clinicians responsible for acting on them